NUS GRIP
Ambient Systems

Assignment #2

September 8, 2020

Assignment 1 Revisited



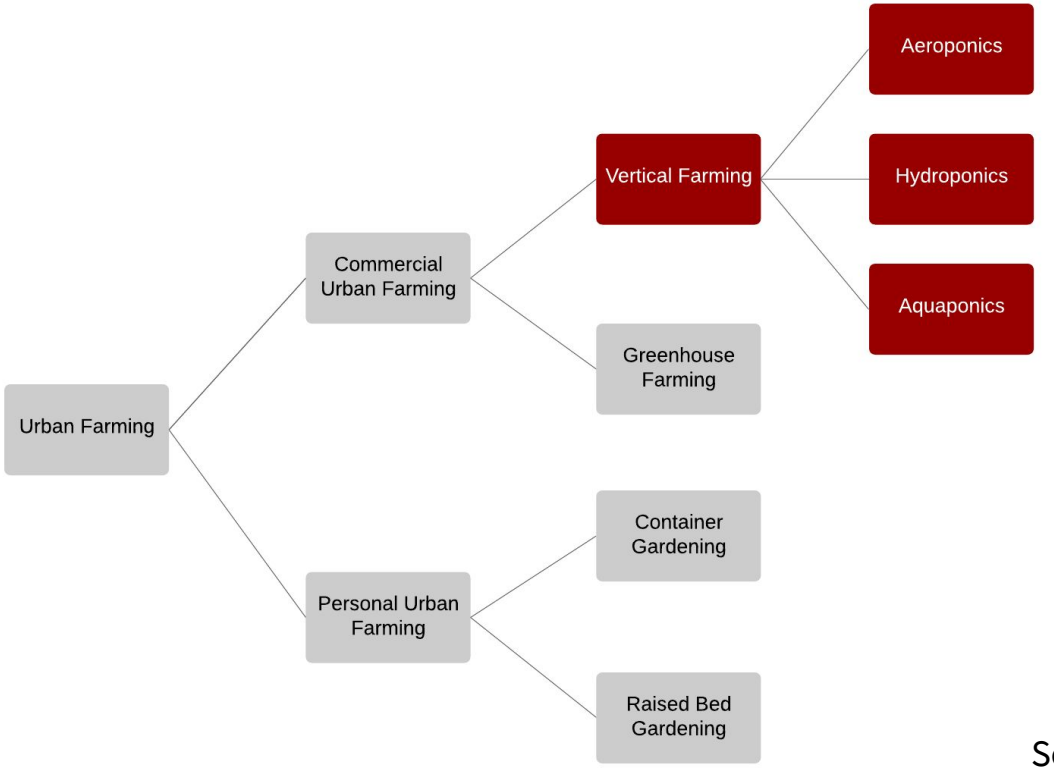
Introduction

Vision: create a safe, healthy, and productive indoor environment for all

Mission: become a leading global provider of building management and automation systems

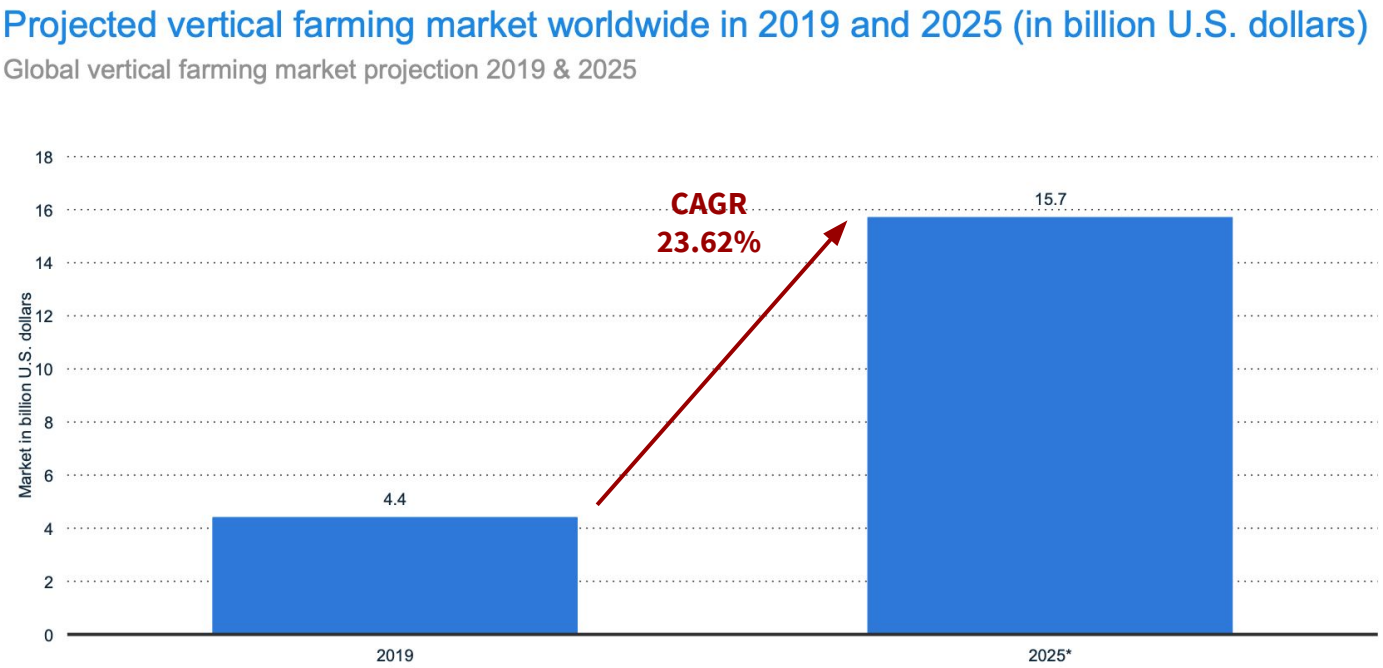
Goal: develop intelligent building automation solutions enabled by IoT and that delight property owners and building operators, starting with the indoor farms segment

Market Segmentation



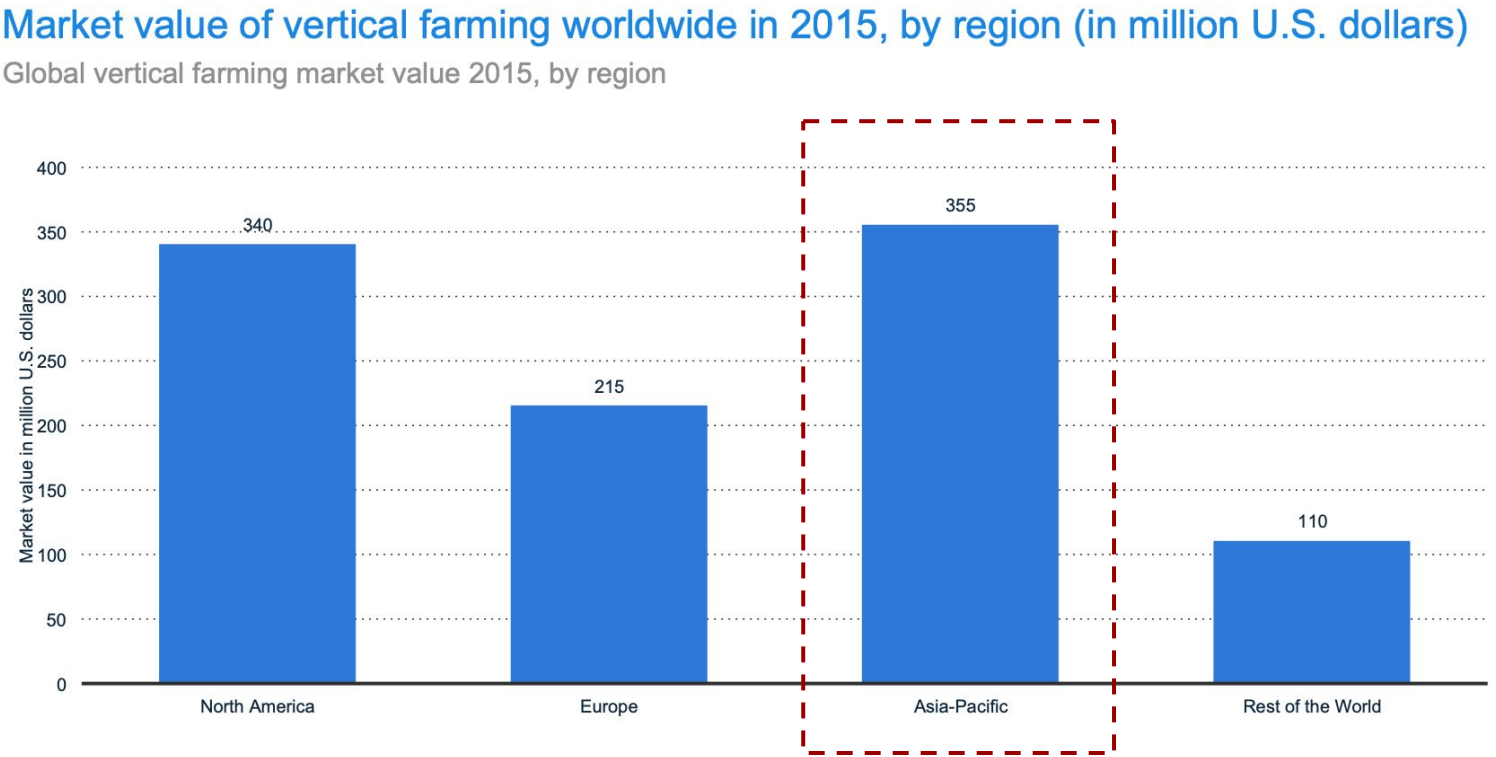
Source: Frost & Sullivan

Indoor Farming Market

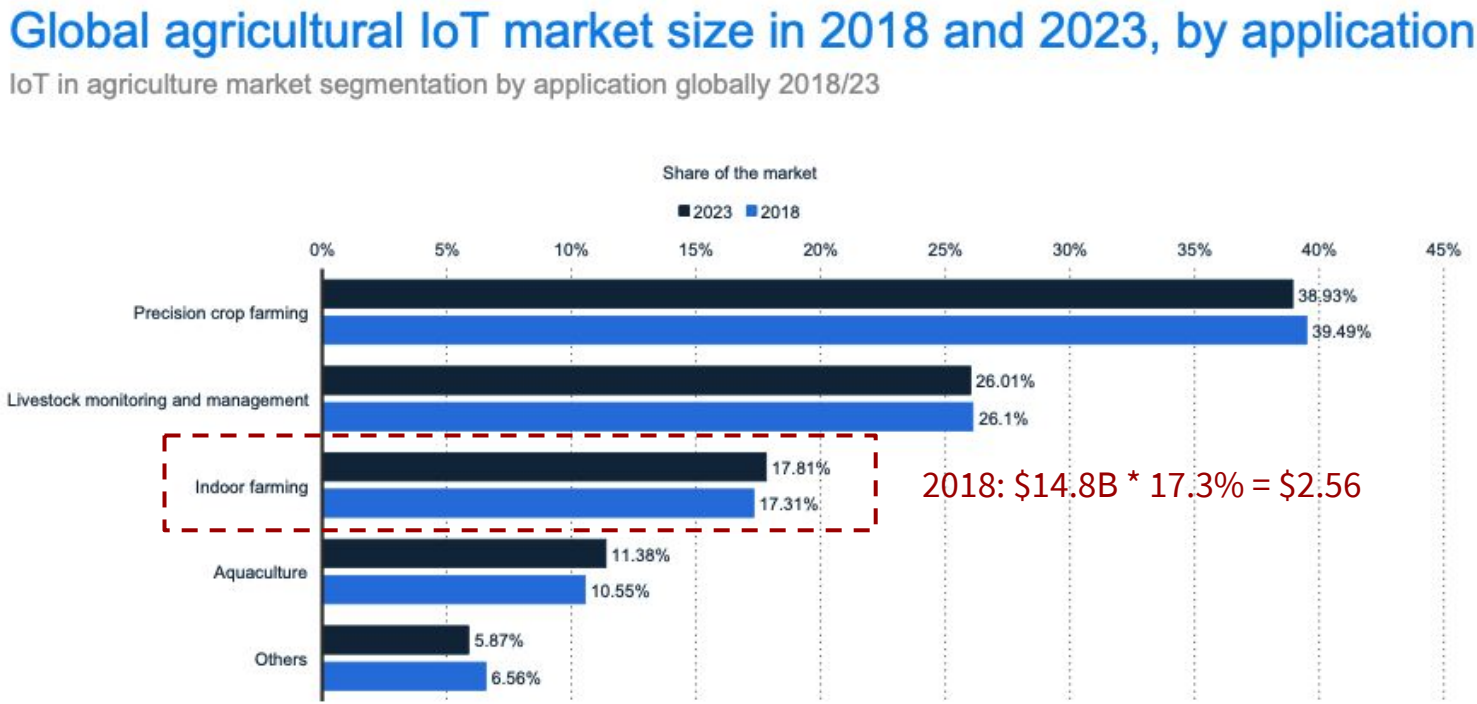


Source: Statista

Indoor Farming Market



Agricultural IoT Market



2018: $14.8B * 17.3% = $2.56

Source: Statista

Vertical Farming is a new industry and secondary market research is somewhat incomplete.

We will have to rely on primary market research for further validation.

Section 1: MVP



MVP: “Climate-as-a-Service”

Components:

1. Wireless that measures and manages key parameters
2. Connection with existing systems
3. Dashboard with all parameters

Advantages:

1. Opens up the market (major expense for growers)
2. Minimizes chance of replications
3. Simple streamlined process for the customer

Section 2: Product Specifications



Specifications

Business model: climate-as-a-service

We manage everything for $X / sq.m

Boundary: we do not manage anything that touches the plants (e.g., racks, water, nutrients, pesticide, etc.)

Exact sensor type, model, quantity will depend on the farm and user’s needs

Section 3: Product Roadmap



Product Roadmap



First product milestone, which is also the MVP, will consists of a wireless network of sensors and a platform to display this data in one place.

Second product milestone will include a layer of added intelligence to the existing system in order to streamline operations and potentially enhance production.

Third product milestone will include a layer of controls that will enable fine-tuning of climate parameters remotely and automatically to create optimal conditions.

Thank you

Appendix



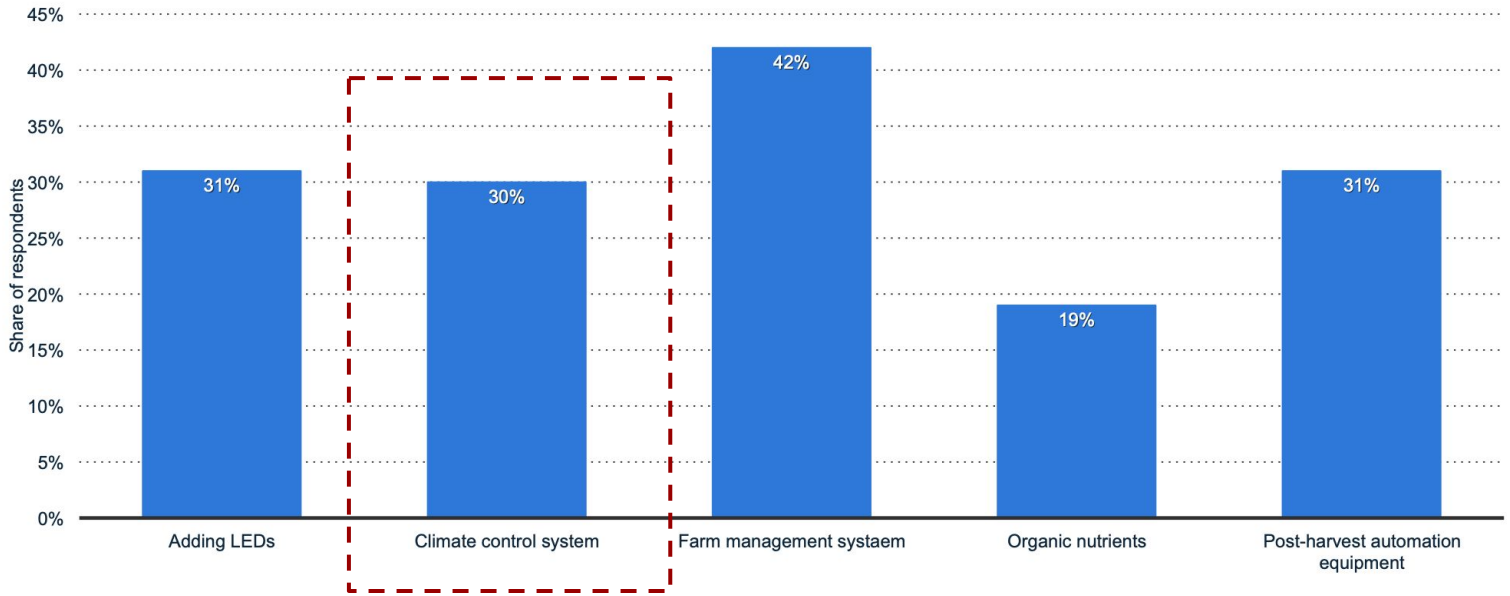
Sample Companies

- https://www.geekwire.com/2019/jeff-bezos-backed-vertical-agriculture-startup-plenty-hibernates-plans-seattle-farm/
- https://www.cropx.com
- http://www.onfarm.com
- https://www.farmobile.com
- https://artemisag.com
- https://upgrownfarming.co/
- www.skygreens.com
- http://www.susteniragriculture.com/
- https://www.life3.co/
- https://www.vertivegies.farm/
- https://www.pfsap.panasonic.com.sg/ps_client_agriculture.aspx

Market

Intended technology and farm improvements in indoor farming in the next year worldwide as of 2016

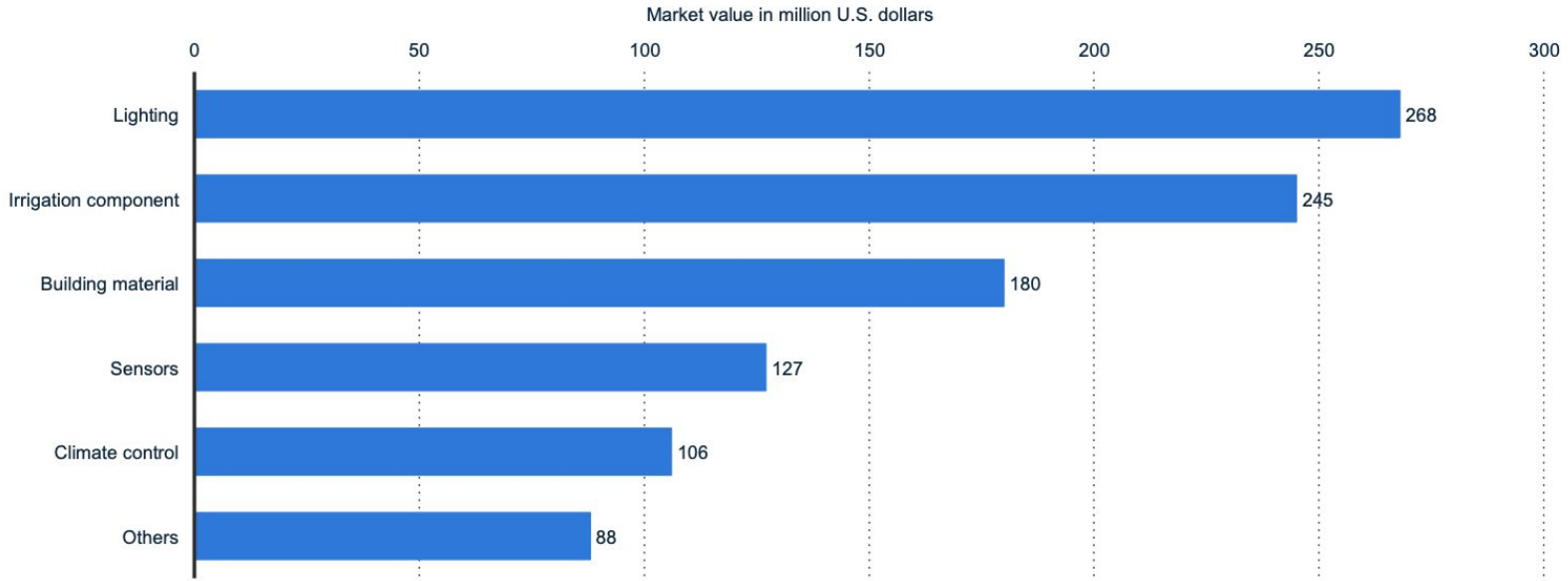
Planned improvements in indoor farming worldwide 2016



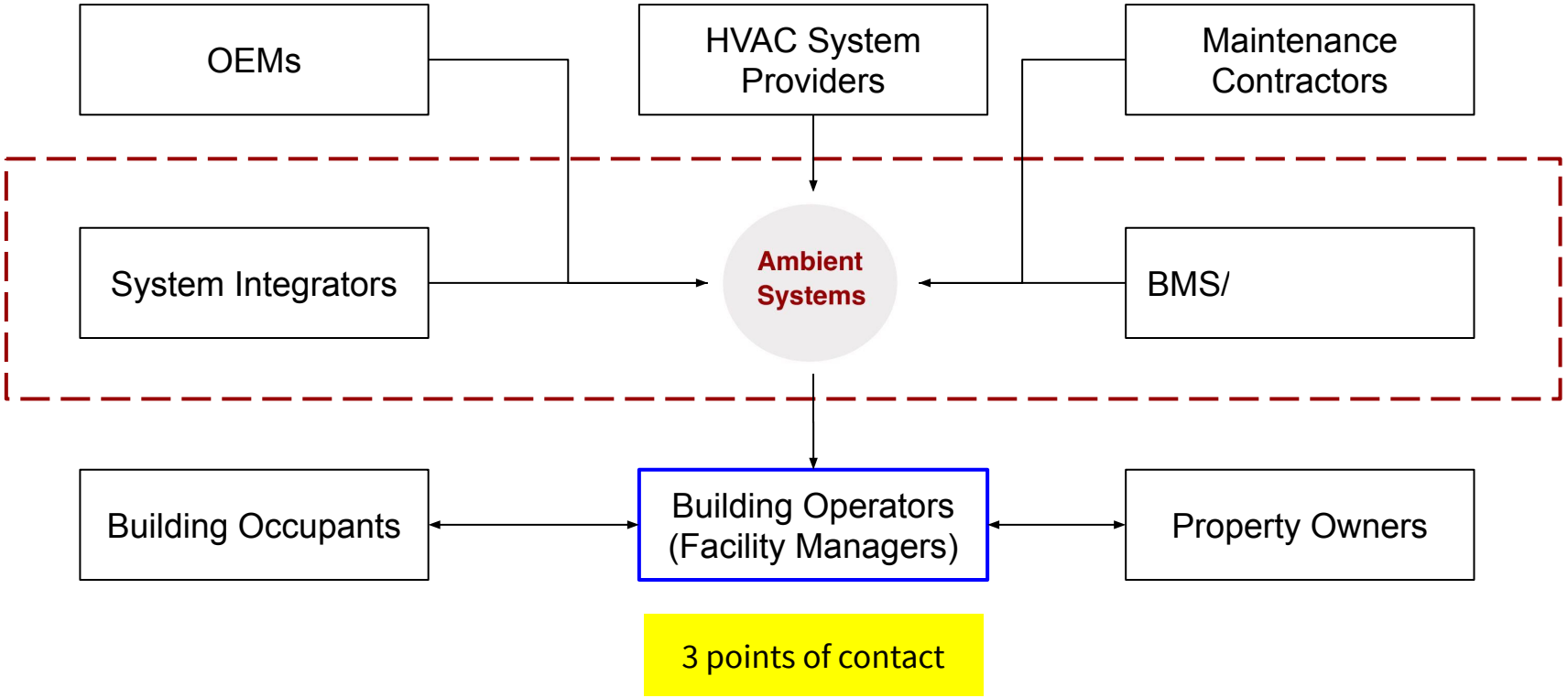
Indoor Farming Market

Revenue of vertical farming worldwide in 2016, by technology (in million U.S. dollars)*

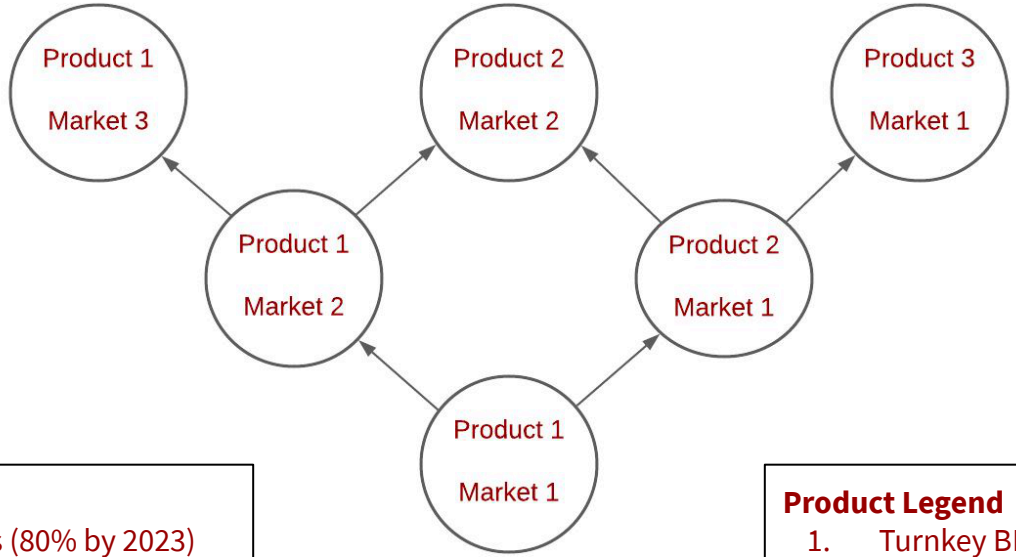
Global vertical farming revenue 2016, by technology



Value Chain Analysis



Bowling Pin Diagram



Market Legend

1. M1: SG Indoor Farms (80% by 2023)
2. M2: APAC C&I (50% by 2029)
3. M3: Europe C&I (50% by 2031)

Product Legend

1. Turnkey BMS (system integrator)
2. enabled BAS
3. enabled

3s Framework

